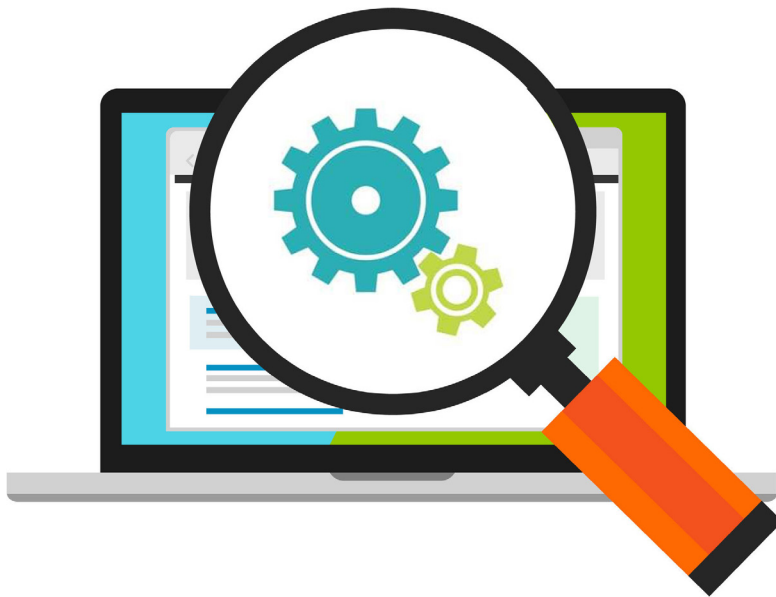


# Testing SaltStack Environment with KitchenSalt, Docker and TestInfra

Have you ever struggled to test SaltStack states or formulas before rolling them out into production, or wondered how to validate SaltStack code changes without breaking the existing SaltStack environment? This tutorial shows how to design a seamless testing and validating workflow using KitchenSalt, Docker and TestInfra to test the SaltStack environment independently.



**S**altStack or Salt is one of the most potent configuration management tools to automate infrastructure at scale.

It facilitates users to efficiently manage the end-to-end life cycle of the servers, starting from bootstrapping a new node to deploying a live application on it. Despite being a robust automation engine, there is no easy way to test and validate the Salt code changes other than running them on individual servers and checking the results manually. Doesn't sound like an ideal practice, especially when dealing with a massive infrastructure and considering a bunch of developers working simultaneously on the same Salt code. Moreover, making such changes in production without independent testing could lead to a non-functioning Salt infrastructure.

But KitchenSalt and TestInfra reduce that risk. They allow users to replicate an independent infrastructure that is similar to the production environment locally and execute end-to-end validations against it.

What follows in this article assumes that you have the following knowledge:

- Understanding of SaltStack and its basic building blocks
- Understanding of Docker and its core commands
- Basics of Python & Ruby (Gem) for installing dependent libraries

You will need the following in order to design the workflow: